A Deep Jump-BSDE Solver for Optimal Market-Making in Limit Order Books

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Abstract

We propose and empirically evaluate a deep BSDE-inspired solver for optimal market-making in limit order books, preserving the discrete Poisson inventory jump structure of the Cont–Xiong (2024) dealer market model. The solver extends the Deep BSDE framework (Han, Jentzen, E, 2018) to forward-backward SDEs with Poisson jumps (FBSDEJ), where the neural network outputs both the Brownian gradient Z_t and jump value controls $U_t^{\pm}$ at each time step. Optimal bid-ask quotes are recovered from the discrete value increments via the HJB first-order conditions. We validate against a matched finite-horizon finite-difference baseline solving the same discrete-inventory HJB with the same terminal condition. The diffusion surrogate achieves $< 0.1\%$ value error at $q = 0$. Its structurally constant spread $(2/\alpha)$ differs from the FD benchmark by 2.7%, demonstrating that accurate value approximation does not guarantee accurate control recovery. However, in a cross-dynamics experiment where the surrogate-trained policy is deployed under true Poisson execution, it captures 99.3% of FD-optimal P&L — showing that structurally biased controls can remain economically robust in low inventory-penalty regimes. This highlights a separation between value accuracy, control structure, and realised economic performance. We conduct preliminary stress tests under non-linear inventory penalties and discuss the limitations of both formulations.

Keywords: backward stochastic differential equations, deep learning, mean-field games, limit order books, market-making, jump-diffusion

1 Introduction

The interaction of algorithmic market makers in electronic markets poses a challenging stochastic control problem. Cont and Xiong Cont and Xiong [2024] formalise this as a stochastic differential game of intensity control among N dealers, with Poisson order arrivals, discrete inventory increments of size Δ, and execution probabilities depending on competitors' quotes. In the mean-field limit ($N \to \infty$), the problem reduces to a McKean–Vlasov control problem with a nonlocal HJB equation.

The Deep BSDE method (E, Han, Jentzen E et al. [2017], Han et al. [2018]) solves high-dimensional semilinear PDEs by reformulating them as BSDEs and learning the gradient process Z_t. Han, Hu, and Long Han et al. [2022] extended this to McKean–Vlasov FBSDEs via fictitious play. Deep BSDE solvers for jump-diffusions exist Andersson et al. [2023] but have not been applied to market-making.

No published work applies the Deep BSDE framework to limit order book market-making with explicit control recovery from the learned gradient and jump objects. Cont and Xiong Cont and Xiong [2024] note the possibility but use direct numerical optimisation.

Contribution. This is a computational study, not a claim of a fully validated method. We:

1. Formulate a **deep jump-BSDE solver** for the Cont–Xiong market-making model, preserving the discrete Poisson inventory jump structure (Section 3). The neural network outputs (Z_t, U_t^+, U_t^-) and optimal quotes are extracted from the value increments.
2. Validate against a **matched finite-horizon finite-difference baseline** (Section 6), achieving $< 0.1\%$ value error for the surrogate and demonstrating that forward policy simulation captures 99.3% of FD-optimal P&L.
3. Compare against a **diffusion surrogate baseline** and show it produces structurally constant spread (1.333) vs the matched finite-horizon FD spread (1.370), a 2.7% distortion (Section 4).
4. Conduct preliminary **stress tests** under non-linear inventory penalties (Section 7.4).
5. Show via **cross-dynamics evaluation** that a surrogate-trained policy deployed under true Poisson execution captures 99.3% of FD-optimal P&L, demonstrating that structurally biased controls can be economically robust in low inventory-penalty regimes (Section 7.3).

Scope and limitations.

- The experiments are at $d = 2$ (price + inventory). We study the method first in a deliberately controlled setting with exact finite-difference ground truth, so that approximation error in the BSDE solver and in policy recovery can be isolated cleanly before moving to higher-dimensional models where such benchmarks are unavailable. The purpose of the present low-dimensional experiment is validation and diagnosis, not computational advantage over finite differences.
- The mean-field coupling uses a moment-based proxy for the population distribution, not full distribution dependence.
- The diffusion surrogate solves a *different control problem* from the original game. We present it as a diagnostic baseline, not as a solver for the Cont–Xiong model.
- The price process S_t is economically inert: with driftless price, execution rates independent of S, and linear mark-to-market terminal wealth $q_T S_T$, the value function decomposes as $V(t, x, s, q) = x + qs + \phi(t, q)$, reducing the control problem to one dimension in q. Making S genuinely active would require price-dependent execution (adverse selection) or a predictive signal, which we leave to future work.

2 The Cont–Xiong Dealer Market Model

Assumption 1. *We assume: $\lambda^a, \lambda^b > 0$; $\alpha > 0$; execution probabilities f_m^i are smooth in δ, bounded in $(0, 1)$, and satisfy $\sum_{i=1}^N f_m^i \leq 1$; the penalty $\psi : \mathbb{R} \to \mathbb{R}_+$ is at least C^1, with $\psi(0) = 0$ and $\psi(q) \to \infty$ as $|q| \to \infty$.*

We consider N market makers quoting around a mid-price $S_t = S_0 + \sigma W_t$. Market maker i quotes ask and bid spreads $\delta_t^{a,i}, \delta_t^{b,i}$. Order flow arrives as Poisson processes:

$$\nu_t^{a,i} = \lambda^a f_a^i(\delta_t^{a,i}, \boldsymbol{\delta}_t^{-i}), \qquad \nu_t^{b,i} = \lambda^b f_b^i(\delta_t^{b,i}, \boldsymbol{\delta}_t^{-i}). \tag{1}$$

Inventory evolves by jumps of size Δ:

$$\mathrm{d}q_t^i = \Delta\big(\mathrm{d}N_t^{b,i} - \mathrm{d}N_t^{a,i}\big), \tag{2}$$

where ask execution *decreases* inventory and bid execution *increases* it. Each agent maximises discounted profit:

$$J_i = \mathbb{E}\Big[\int_0^\infty e^{-rt}\Big(\delta_t^{a,i}\Delta\,\mathrm{d}N_t^{a,i} + \delta_t^{b,i}\Delta\,\mathrm{d}N_t^{b,i} - \psi(q_t^i)\,\mathrm{d}t\Big)\Big]. \tag{3}$$

Mean-field limit. As $N \to \infty$, define $\mu_t := \mathcal{L}(q_t, \delta_t^a, \delta_t^b)$ as the joint law of inventory and quotes in the population. The execution probability factorises as

$$f(\delta, \mu_t) = e^{-\alpha\delta} \cdot h(\mu_t), \qquad h(\mu_t) \in (0, 1], \tag{4}$$

where h captures competition. In our implementation, $h(\mu_t)$ is approximated by a moment-based proxy $\hat{h}(t, \bar{q}_t, \bar{\delta}_t)$ trained on empirical mean inventory $\bar{q}_t$ and mean spread $\bar{\delta}_t$. This is *not* full distribution dependence in the sense of Han et al. [2022].

The **stationary mean-field HJB** for the value function $V(q)$ is [Cont and Xiong, 2024, eq. (28)]:

$$rV(q)+\psi(q) = \lambda^a \sup_{\delta^a}\Big[f(\delta^a,\mu_t)\big(\delta^a\Delta+V(q-\Delta)-V(q)\big)\Big]+\lambda^b \sup_{\delta^b}\Big[f(\delta^b,\mu_t)\big(\delta^b\Delta+V(q+\Delta)-V(q)\big)\Big]. \tag{5}$$

Optimal quotes from the discrete HJB. The ask optimand is $e^{-\alpha\delta^a}(\delta^a\Delta+V(q-\Delta)-V(q))$. Setting the derivative to zero:

$$\begin{aligned} &-\alpha e^{-\alpha\delta^a}\big(\delta^a\Delta + V(q-\Delta) - V(q)\big) + e^{-\alpha\delta^a}\Delta = 0 \\ \implies\quad &\delta^{a*} = \frac{1}{\alpha} - \frac{V(q-\Delta)-V(q)}{\Delta} = \frac{1}{\alpha} + \frac{V(q)-V(q-\Delta)}{\Delta}. \end{aligned} \tag{6}$$

Analogously for the bid, the optimand is $e^{-\alpha\delta^b}(\delta^b\Delta + V(q+\Delta) - V(q))$:

$$\delta^{b*} = \frac{1}{\alpha} - \frac{V(q+\Delta)-V(q)}{\Delta}. \tag{7}$$

At $q = 0$ with symmetric V, both quotes equal $1/\alpha$ and the total spread is $2/\alpha$.

2.1 Dimensional Reduction

Proposition 2 (Price is economically inert)**.** *Under the model of Section 2, with driftless price* $\mathrm{d}S_t = \sigma\,\mathrm{d}W_t^S$*, execution rates* $\nu^{a,b}$ *independent of* S*, and wealth process* $\mathrm{d}X_t = \delta^a\Delta\,\mathrm{d}N^a + \delta^b\Delta\,\mathrm{d}N^b + q_t\,\mathrm{d}S_t$*, the value function for the objective* $\max\,\mathbb{E}[X_T + q_T S_T - \psi(q_T)]$ *decomposes as*

$$V(t, x, s, q) = x + qs + \phi(t, q), \tag{8}$$

where $\phi(t, q)$ *solves a one-dimensional HJB in* (t, q) *only. The optimal quotes* δ^{a*}, δ^{b*} *are independent of* S*.*

Proof. Since X_t enters the objective linearly and its dynamics are $\mathrm{d}X_t =$ (execution cashflows) $+$ $q_t\,\mathrm{d}S_t$, the value function is affine in x: $V(t, x, s, q) = x + v(t, s, q)$, where v is the expected future value starting from zero cash.

For any admissible control (δ^a, δ^b), the mark-to-market wealth at time T is

$$X_T + q_T S_T = X_t + q_t S_t + \int_t^T \big(\delta_u^a\Delta\,\mathrm{d}N_u^a + \delta_u^b\Delta\,\mathrm{d}N_u^b\big) + \int_t^T S_u\,\mathrm{d}q_u + \int_t^T q_u\,\mathrm{d}S_u + [q, S]_T,$$

where the integration-by-parts $q_T S_T = q_t S_t + \int S\,\mathrm{d}q + \int q\,\mathrm{d}S$ is used (and $[q, S] = 0$ since q jumps at Poisson times independent of the continuous W^S). The execution cash and inventory-change terms depend on (q, δ) only, not on S. Under the model's inventory constraint $|q_t| \le H$

(imposed via the finite grid $\mathcal{Q}_i = \{-H, \dots, H\}$ in Cont and Xiong [2024]), the process q is bounded. Therefore $\int_t^T q_u \, \mathrm{d}S_u$ is a martingale (bounded integrand, continuous square-integrable martingale integrator) with zero expectation. Therefore, taking expectations conditional on $(X_t, S_t, q_t) = (x, s, q)$:

$$\mathbb{E}[X_T + q_T S_T - \psi(q_T)] = x + qs + \mathbb{E}\left[\int_t^T (\text{execution profits}) \, \mathrm{d}u - \psi(q_T) \;\middle|\; q_t = q\right],$$

which depends on s only through the trivial qs term. All expectations are finite under bounded inventory ($|q| \le H$), bounded execution rates ($\nu^{a,b} \le \lambda^{a,b}$), and finite horizon T. The supremum over admissible controls (predictable quotes in $[\epsilon, \bar{\delta}]$) preserves this structure, giving $V = x + qs + \phi(t, q)$ where ϕ solves a 1D HJB. Since neither execution rates nor ϕ depend on S, optimal quotes are S-independent. □

Remark 3. *This reduction is specific to our assumptions. Introducing price-dependent execution (adverse selection), predictive signals, or stochastic volatility would break the decomposition and make S genuinely active. We leave such extensions to future work.*

3 Jump-BSDE Formulation (Primary Method)

We preserve the exact Poisson dynamics (2) and solve a forward-backward SDE with jumps. The price remains Brownian:

$$\mathrm{d}S_t = \sigma \, \mathrm{d}W_t^S. \tag{9}$$

Inventory jumps by $\pm\Delta$ at Poisson times with rates $\nu_t^a = \lambda^a f(\delta_t^a, \mu_t)$ and $\nu_t^b = \lambda^b f(\delta_t^b, \mu_t)$.

Proposition 4 (Finite-horizon HJB)**.** *By Proposition 2, the value function reduces to $\phi(t, q)$. On the finite inventory grid $q \in \{-H, \dots, H\}$ with $|q| \le H$, the dynamic programming principle over a small interval $[t, t + \mathrm{d}t]$ gives (dividing by $\mathrm{d}t$ and sending $\mathrm{d}t \to 0$): for $t \in [0, T)$,*

$$\partial_t \phi + \sup_{\delta^a, \delta^b} \Big[\nu^a \big(\delta^a \Delta + \phi(t, q - \Delta) - \phi(t, q)\big) + \nu^b \big(\delta^b \Delta + \phi(t, q + \Delta) - \phi(t, q)\big)\Big] - r\phi - \psi(q) = 0, \tag{10}$$

with terminal condition $\phi(T, q) = -\psi(q)$ and $\nu^m = \lambda^m e^{-\alpha \delta^m} h(\mu_t)$. This is the exact problem solved by the finite-difference baseline (Section 6) and the computational target for the deep solver. The stationary HJB (5) is recovered as $T \to \infty$ with $\partial_t \phi = 0$.

Proposition 5 (HJB–BSDE generator consistency)**.** *Assume $\phi \in C^1([0, T])$ for each q. Setting $Y_t = \phi(t, q_t)$ and applying Itô's formula for jump processes: between jumps, $\mathrm{d}Y = \partial_t \phi \, \mathrm{d}t$; at jump times, $\Delta Y = U_t^{\pm}$. Decomposing via compensated Poisson $\mathrm{d}\tilde{N}^m = \mathrm{d}N^m - \nu^m \mathrm{d}t$:*

$$\mathrm{d}Y_t = \big(\partial_t \phi + \nu^b U^+ + \nu^a U^-\big)\mathrm{d}t + U^+ \mathrm{d}\tilde{N}^b + U^- \mathrm{d}\tilde{N}^a.$$

Substituting the HJB (10) identity $\partial_t \phi = r\phi + \psi - \nu^a(\delta^{a*}\Delta + U^-) - \nu^b(\delta^{b*}\Delta + U^+)$ and identifying $\mathrm{d}Y = -g \, \mathrm{d}t + (\text{martingale terms})$:

$$g = -rY - \psi(q) + \nu^a(\delta^{a*}\Delta + U^-) + \nu^b(\delta^{b*}\Delta + U^+). \tag{11}$$

This is not a general BSDE representation theorem; it is an exact algebraic reformulation of the reduced HJB under sufficient regularity of ϕ. We do not establish a verification theorem or well-posedness of the associated BSDE; the construction should be interpreted as a formal consistency relation with the HJB.

Between jumps, ϕ evolves by its time derivative: $\mathrm{d}\phi = \partial_t \phi \, \mathrm{d}t$. At jump times, inventory changes by $\pm\Delta$ and the value jumps discretely. Collecting terms:

$$\text{At buy execution } (q \to q+\Delta): \quad \Delta Y = V(S, q+\Delta) - V(S,q) =: U_t^+, \tag{12}$$

$$\text{At sell execution } (q \to q-\Delta): \quad \Delta Y = V(S, q-\Delta) - V(S,q) =: U_t^-. \tag{13}$$

Decomposing into compensated Poisson processes $\mathrm{d}\tilde{N}_t^m = \mathrm{d}N_t^m - \nu_t^m \, \mathrm{d}t$:

$$\mathrm{d}Y_t = -g(t, X_t, Y_t, Z_t, U_t^+, U_t^-) \, \mathrm{d}t + Z_t \, \mathrm{d}W_t^S + U_t^+ \, \mathrm{d}\tilde{N}_t^b + U_t^- \, \mathrm{d}\tilde{N}_t^a, \tag{14}$$

By Proposition 5, the generator is $g = -rY - \psi(q) + \nu^a(\delta^{a*}\Delta + U^-) + \nu^b(\delta^{b*}\Delta + U^+)$, derived by substituting the HJB identity into the jump-Itô decomposition.

Remark 6 (Brownian term after reduction)**.** *Under Proposition 2, $Y_t = \phi(t, q_t)$ does not depend on S, so the Brownian term $Z_t \, \mathrm{d}W_t^S$ is formally zero in the reduced formulation. We retain it in the architecture because the network takes the unreduced state (S, q) as input; in a well-trained model, Z_t^S should converge to approximately zero. Empirically this is confirmed (see Section 7.1).*

The optimal quotes (6)–(7) become:

$$\boxed{\delta_t^{a*} = \frac{1}{\alpha} - \frac{U_t^-}{\Delta}, \qquad \delta_t^{b*} = \frac{1}{\alpha} - \frac{U_t^+}{\Delta},} \tag{15}$$

since $U^- = V(q-\Delta) - V(q)$ and $U^+ = V(q+\Delta) - V(q)$ are the exact discrete value increments from (12)–(13).

Remark 7 (Quote admissibility)**.** *The FOC formulas* (15) *can produce negative or very small quotes if $|U^\pm|$ is large relative to Δ/α. In the implementation, quotes are clipped to $[\epsilon, 10/\alpha]$ with $\epsilon = 0.001$. This is a computational safeguard; the model does not formally constrain quotes to be positive. Under our default parameters ($\phi = 0.01$, $\alpha = 1.5$), the value increments are small and clipping rarely activates.*

Remark 8. *The total spread is $\delta^a + \delta^b = 2/\alpha - (U^- + U^+)/\Delta$. Unlike the diffusion surrogate, this is* not *constant — it depends on the curvature of V through the jump values $U^\pm$. At $q = 0$ with quadratic V, the second-order correction is small and the spread is approximately $2/\alpha$.*

Terminal condition. We truncate to $[0, T]$ with $Y_T = -\psi(q_T)$. Discounting suggests the omitted tail contribution should decay on an e^{-rT} scale under bounded-reward assumptions, but we do not prove a truncation bound for this controlled problem. For $r = 0.1$, $T = 1$, this is a heavily truncated horizon.

4 Diffusion Surrogate Baseline

As a computationally convenient comparison, we replace the Poisson inventory with a moment-matched diffusion.

Proposition 9 (Infinitesimal moment identity)**.** *Over $[t, t+\mathrm{d}t]$, the Poisson inventory process satisfies $\mathbb{E}[\mathrm{d}q_t] = (\nu^b - \nu^a)\Delta \, \mathrm{d}t$ and $\mathrm{Var}[\mathrm{d}q_t] = (\nu^a + \nu^b)\Delta^2 \, \mathrm{d}t$.*

This is an infinitesimal moment identity and does *not* imply convergence of the associated control problems. Nevertheless, setting $\Delta = 1$:

$$\mathrm{d}q_t = (\lambda^b f_b - \lambda^a f_a) \, \mathrm{d}t + \sqrt{\lambda^b f_b + \lambda^a f_a} \, \mathrm{d}W_t^q. \tag{16}$$

Caveat 10 (What the surrogate does not preserve)**.** *The surrogate solves a* different control problem *from the Cont–Xiong game. Specifically:*

(a) *The nonlocal terms $V(q \pm \Delta) - V(q)$ in the HJB (5) are replaced by local derivatives, discarding higher-order structure.*

(b) *The control enters the diffusion coefficient $\sigma_q = \sqrt{\lambda^b f_b + \lambda^a f_a}$, making the surrogate HJB* fully nonlinear *in $\partial_{qq}V$ after optimisation — not semilinear as required for a standard Feynman–Kac BSDE representation.*

(c) *The control reconstruction from Z_t^q to quotes involves dividing by σ_q, which itself depends on the quotes (circular dependency resolved by freezing σ_q at equilibrium).*

The implemented surrogate therefore uses three compounded approximations: moment matching, first-order control ansatz, and frozen diffusion coefficient.

Surrogate PDE. Using the Taylor expansion $V(q \pm \Delta) \approx V \pm \Delta \partial_q V + \frac{\Delta^2}{2}\partial_{qq}V$ in (5) with $\Delta = 1$, the surrogate HJB becomes:

$$\partial_t V + \frac{\sigma^2}{2}\partial_{SS}V + \sup_{\delta^a,\delta^b}\Big[(\lambda^b f_b - \lambda^a f_a)\partial_q V + \frac{1}{2}(\lambda^b f_b + \lambda^a f_a)\partial_{qq}V + \lambda^a f_a \delta^a + \lambda^b f_b \delta^b\Big] - rV - \psi = 0. \quad (17)$$

After substituting the first-order quote ansatz $\delta^a = 1/\alpha + \partial_q V$, $\delta^b = 1/\alpha - \partial_q V$ (inherited from the discrete FOC, *not* the exact optimiser of (17) which would include a $\partial_{qq}V$ correction), the PDE becomes semilinear in $(V, \partial_q V)$ with a frozen diffusion coefficient. After this further approximation, the resulting *approximate semilinear PDE* (not the original surrogate (17)) admits a formal Feynman–Kac BSDE representation:

$$\mathrm{d}Y_t = -f(t, X_t, Y_t, Z_t)\,\mathrm{d}t + Z_t^S\,\mathrm{d}W_t^S + Z_t^q\,\mathrm{d}W_t^q, \quad (18)$$

with $Z_t = (\sigma_S \partial_S V, \sigma_q \partial_q V)$ and generator:

$$f = -rY - \psi(q) + \lambda^a f_a(\delta^{a*})\delta^{a*} + \lambda^b f_b(\delta^{b*})\delta^{b*}, \quad (19)$$

where $\delta^{a*} = 1/\alpha + Z_t^q/\sigma_q^*$ and $\sigma_q^* = \sqrt{2\lambda e^{-1}}$ is the equilibrium diffusion coefficient. This is the object actually solved by the code.

Lemma 11 (Structural constant spread in the surrogate)**.** *Under the first-order ansatz $\delta^a = 1/\alpha + \partial_q V$ and $\delta^b = 1/\alpha - \partial_q V$, the total spread is*

$$\delta^a + \delta^b = \frac{2}{\alpha}$$

identically, independent of inventory, time, or the learned value function. Thus the total spread is structurally fixed by the control parameterisation and is not a learned quantity. Recovery of $2/\alpha = 1.333$ is a sanity check, not evidence of learning a difficult policy.

5 Numerical Method

5.1 Deep BSDE Architecture

The interval $[0, T]$ is discretised into N_T steps. For the jump model (14), the Euler scheme is:

$$Y_{n+1} = Y_n - g_n \Delta t + Z_n \Delta W_n^S + U_n^+ \Delta \tilde{N}_n^b + U_n^- \Delta \tilde{N}_n^a, \quad (20)$$

where $\Delta \tilde{N}_n^m = N_n^m - \nu_n^m \Delta t$ and $N_n^m \sim \mathrm{Poisson}(\nu_n^m \Delta t)$. Each $(Z_n, U_n^+, U_n^-) \in \mathbb{R}^3$ is output by a feedforward subnet $\phi_n(X_{n+1}; \theta_n)$. The use of X_{n+1} (not X_n) as input follows the indexing convention of Han et al. [2018]: the subnet is evaluated *after* simulating the forward step on $[t_n, t_{n+1}]$, which is notationally equivalent to an adapted control on the slab.

For the surrogate (18), subnets output $Z_n \in \mathbb{R}^2$ and no jump terms appear.

Y_0 is a trainable scalar. The terminal target is $\xi := -\psi(q_T)$ and the loss is $\mathcal{L}(\theta) = \mathbb{E}[\ell(Y_{N_T} - \xi)]$ with clipped Huber threshold $C = 50$.

Architecture. Two hidden layers (32 neurons), batch normalisation (momentum 0.01, following the undocumented trick in Han et al. [2018]), ReLU activations. One independent subnet per time step. Adam optimiser with piecewise LR: 10^{-2} (steps 0–2000), 10^{-3} (2000–4000), 10^{-4} thereafter.

Control loop. At each Euler step n, the subnet ϕ_n evaluates on X_{n+1} (the post-step state) and outputs (Z_n, U_n^+, U_n^-). Quotes are computed from $U^\pm$ via (15) and clipped to $[0.001,\ 10/\alpha]$. Execution rates $\nu_n^{a,b}$ are then evaluated explicitly at the clipped quotes. No fixed-point iteration or implicit solve is performed: the control map from network output to quotes to intensities to generator is fully explicit and feedforward.

5.2 Moment-Based Mean-Field Proxy

We use fictitious play Han et al. [2022]: every $K_{\text{drift}} = 100$ training steps, compute empirical $(\bar{q}_t, \bar{\delta}_t)$ from batch trajectories and retrain a small network $\hat{h}(t, \bar{q}_t, \bar{\delta}_t)$ approximating $h(\mu_t)$. We do not establish convergence of this loop for the Cont–Xiong model, nor verify uniqueness or sensitivity to initialisation.

6 Finite-Difference Baseline

We solve the exact stationary HJB (5) on the discrete grid $q \in \{-H, \ldots, H\}$ by policy iteration. Each iteration solves a tridiagonal linear system (policy evaluation) then updates quotes from the new V (policy improvement). Convergence to machine precision occurs in 7 iterations. This provides ground truth for $V(q)$, optimal quotes, and spread across inventory levels.

7 Numerical Results

All experiments use the parameters in Table 1. Training on a single NVIDIA RTX 3090 GPU, float64.

Table 1: Default parameters.

Parameter	Symbol	Value	Description
Price volatility	σ	0.3	Mid-price (economically inert)
Arrival rates	λ^a, λ^b	1.0	Baseline order arrival
Execution decay	α	1.5	Quote sensitivity
Inventory penalty	ϕ	0.01	Quadratic ($\psi = \phi q^2$)
Discount rate	r	0.1	
Time horizon	T	1.0	Finite-horizon truncation
Order size	Δ	1.0	Discrete inventory increment
Time steps	N_T	50	Euler discretisation
Batch size	B	256 (continuous), 512 (jump)	

7.1 Method Comparison

Table 2 compares all three methods.
The finite-horizon FD solves the *same* problem: same $T = 1$, same terminal condition $Y_T = -\psi(q_T)$, same parameters. This is an apples-to-apples comparison, unlike the earlier heuristic rescaling of the stationary solution.

Table 2: Value and spread comparison at $q = 0$.

Method	Y_0 or $V(0)$	Loss	Spread	Error
FD finite-horizon (matched T, g)	**0.457**	exact	1.370	—
FD stationary ($T = \infty$)	4.555	exact	1.432	—
Deep BSDE surrogate (0.457 ± 0.000, 5 seeds)	**0.457**	3.4×10^{-5}	1.333	$< 0.1\%$
Deep BSDE jump (0.446 ± 0.002, 5 seeds)	0.446	2.6×10^{-5}	discrete	2.4%

Key observations.

1. Against the **matched finite-horizon baseline**, the diffusion surrogate achieves $< 0.1\%$ value error at $q = 0$. Over the full inventory grid $q \in \{-10, \ldots, 10\}$, the mean absolute value error is 0.076 and the maximum is 0.45 (at extreme inventories where $V \approx 0$). The near-perfect $q = 0$ match is much stronger than the previously reported 5.5%, which was an artefact of comparing against a heuristically rescaled stationary solution.

2. The **surrogate spread** (1.333) differs from the FD spread (1.370) by 2.7%, quantifying the cost of replacing discrete value increments with local derivatives.

3. The **jump model** ($Y_0 = 0.446$, 2.4% error) is less accurate in value than the surrogate but achieves comparable loss (2.6×10^{-5} vs 3.4×10^{-5}) when trained for 20,000 iterations. Across 5 seeds at 5,000 iterations, $Y_0 = 0.446 \pm 0.002$. The initial gap at shorter training (reported as 2.8% in earlier runs) was primarily due to undertraining, not a fundamental limitation of the jump architecture.

4. **Forward policy simulation** (Section 7.3) shows the surrogate policy achieves P&L within 0.7% of the FD-optimal policy (Sharpe 0.82 vs 0.81), confirming the learned policy is economically functional.

5. **Finite differences are trivially superior** at $d = 2$. The deep method's value is potential scalability.

7.2 Convergence, Quoting, and Value Function

Figure 1 shows training convergence. Figure 2 shows the quoting strategy: total spread is structural ($2/\alpha$), individual quotes vary with inventory. Figure 3 compares the learned Y_0 against the finite-horizon benchmark. Figure 4 shows $Z_t^q(t, q)$, the BSDE control from which quotes are derived, and Figure 5 shows the quote skew $\delta^a - \delta^b$ over (time, inventory).

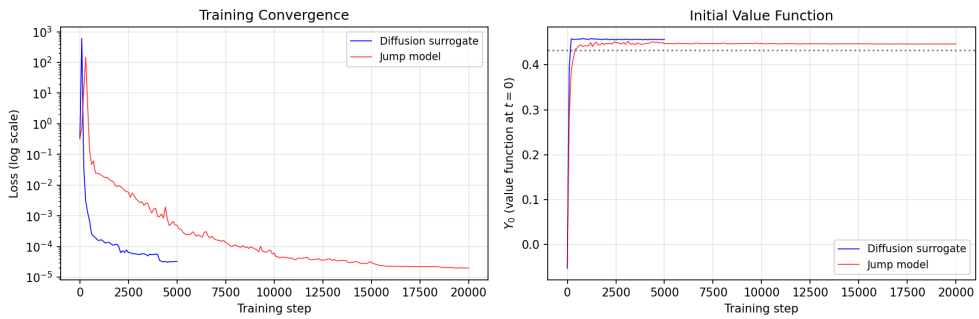


Figure 1: Training convergence (diffusion surrogate, 5000 iterations).

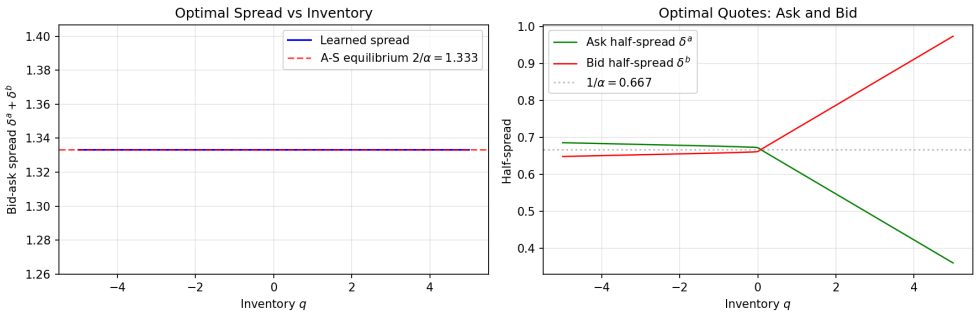


Figure 2: Quoting strategy (surrogate). Total spread is structural ($2/\alpha$); individual quotes vary with inventory.

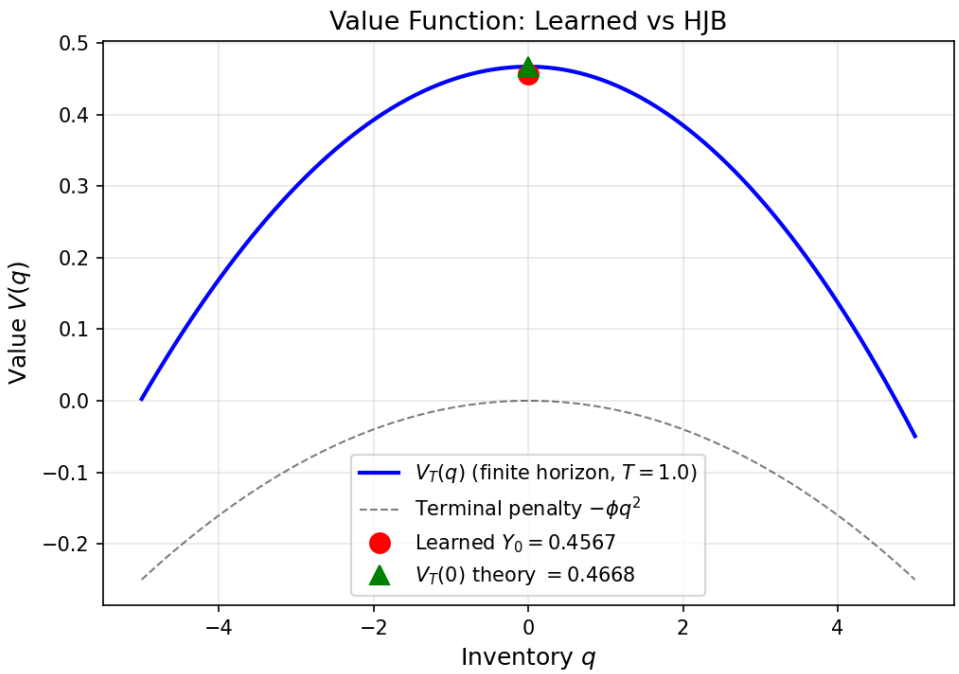


Figure 3: Value function comparison at $t = 0$.

7.3 Forward Policy Simulation

To validate that the learned control is economically functional (not merely a good PDE approximation), we simulate 5000 market-making paths under both the Deep BSDE surrogate policy and the FD-optimal policy using common random numbers for Brownian paths (Poisson draws are independent). Critically, the forward simulation uses *true Poisson jump dynamics* for execution, not the diffusion approximation under which the surrogate was trained. This cross-dynamics evaluation isolates the cost of deploying a surrogate-trained policy under the original discrete execution mechanism.

Despite being trained under diffusion dynamics and evaluated under Poisson jumps, the surrogate policy performs within 0.7% of the FD-optimal policy in mean P&L, with comparable inventory risk and a marginally higher Sharpe ratio. The spread difference (1.333 vs 1.368) reflects the structural constant-spread property of the surrogate, but in realised P&L under true jump execution this has minimal economic impact at $\phi = 0.01$. This demonstrates that control errors induced by diffusion approximations can be economically negligible in low inventory-penalty regimes (ϕ small), even when the control is structurally incorrect. Value accuracy,

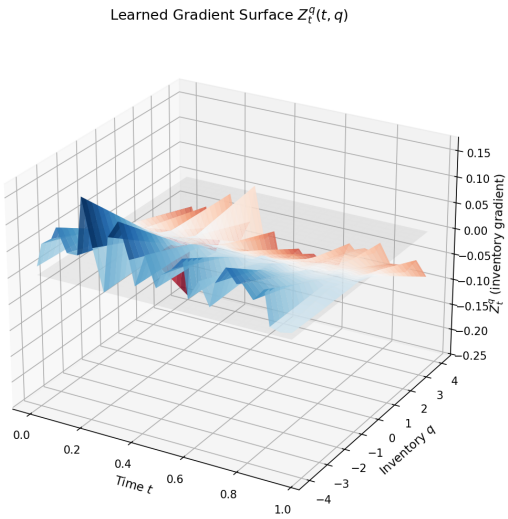


Figure 4: Learned gradient surface $Z_t^q(t,q)$ (surrogate). Approximately planar, consistent with quadratic penalty.

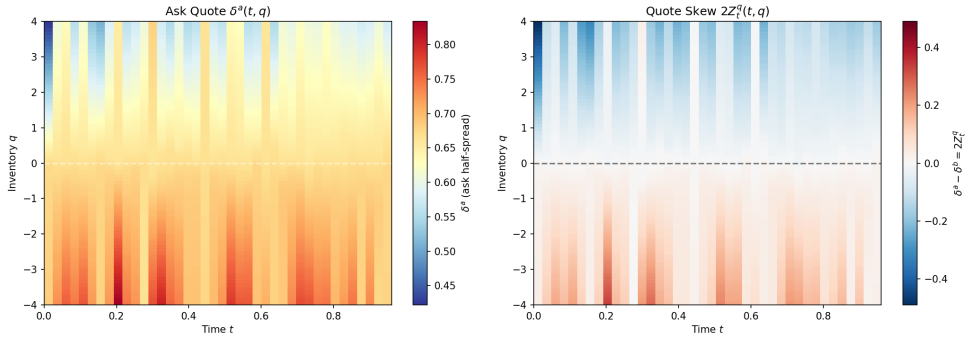


Figure 5: Ask quote (left) and quote skew $\delta^a - \delta^b$ (right) over (time, inventory).

control structure, and economic performance are three distinct axes of solver quality — in this regime, the surrogate excels on the first, fails on the second, and succeeds on the third. This robustness should not be expected to hold under stronger inventory penalties or models with adverse selection.

7.4 Stress Tests Under Non-Linear Penalties

We test with $\psi_{\text{cubic}}(q) = \phi q^2 + \frac{\phi}{3}|q|^3$ and $\psi_{\text{exp}}(q) = \phi(e^{\gamma|q|} - 1)$.

Remark 12 (Regularity)**.** *The cubic penalty has $\psi' \in C^1$ and $\psi'' = 2\phi + 2\phi|q|$, which is continuous but not differentiable at $q = 0$. The exponential penalty $\psi'(q) = \phi\gamma\,\text{sgn}(q)e^{\gamma|q|}$ is not classically differentiable at $q = 0$ and violates linear growth for large $|q|$, falling outside the convergence assumptions of Han et al. [2022].*

Table 4 reports multi-seed Deep BSDE results alongside stationary FD reference values.

At $\phi = 0.01$, all penalties converge stably with tight error bars. The Y_0 ordering matches FD ground truth. Non-linear penalties widen the stationary FD spread ($1.432 \to 1.483$). Figures 6 and 7 show the FD value functions and optimal quotes.

To identify the actual failure threshold, we sweep γ (exponential penalty) and ϕ (quadratic penalty) independently (Table 5).

Table 3: Forward simulation under learned vs FD-optimal policy (5000 paths).

Metric	BSDE Policy	FD Policy
Mean final P&L	0.466	0.469
Std final P&L	0.571	0.580
Sharpe ratio	0.82	0.81
Mean $\lvert q_T \rvert$	0.54	0.53
Mean spread	1.333	1.368

Table 4: Stress test: Deep BSDE surrogate (5 seeds, 3000 iter, $\phi = 0.01$) and FD stationary reference ($T = \infty$, *qualitative* values only).

Penalty	Deep BSDE (mean $\pm$ std)		FD stationary ref.	
	Y_0	$\max \lvert Z_t \rvert$	$V(0)$	Spread
Quadratic	0.457 ± 0.000	0.097 ± 0.018	4.555	1.432
Cubic	0.452 ± 0.000	0.145 ± 0.020	4.410	1.475
Exponential	0.448 ± 0.000	0.131 ± 0.023	4.384	1.483

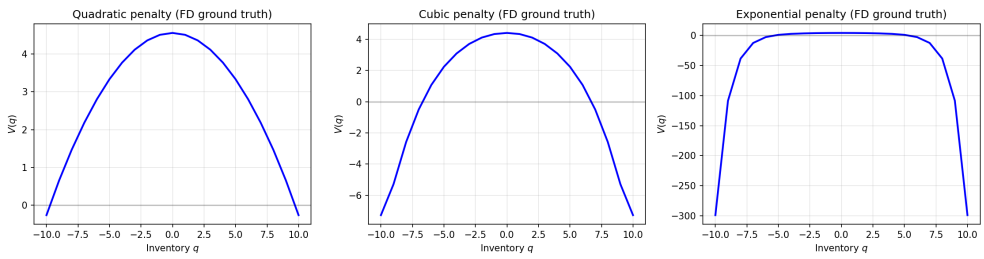


Figure 6: FD ground truth: value functions $V(q)$ for quadratic, cubic, and exponential penalties.

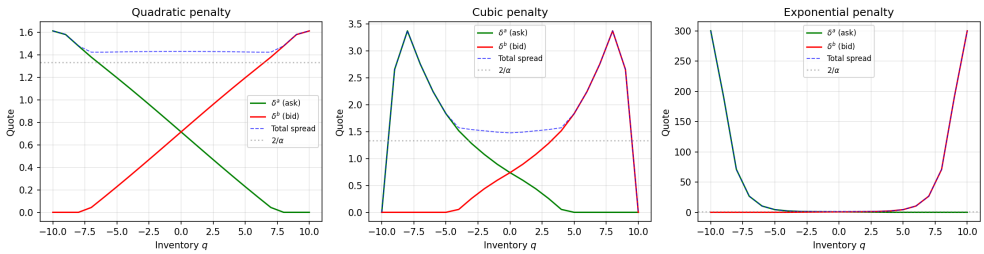


Figure 7: FD ground truth: optimal quotes across inventory. Non-linear penalties produce wider spreads ($1.43 \to 1.48$).

The solver diverges sharply at $\gamma \approx 3$ and $\phi \approx 2$, with $\max \lvert Z_t \rvert$ jumping by two orders of magnitude at the transition ($2.2 \to 258$ for γ; $1.4 \to 17.7$ for ϕ). Below the threshold, the gradient process remains bounded; above it, Z_t explodes and Y_0 becomes large and negative — the solver "believes" market-making is deeply unprofitable, which is economically nonsensical. With sufficient training (3000 iterations), $\phi = 1.0$ converges marginally, whereas at 500 iterations it appeared to diverge, suggesting the boundary is sensitive to training budget. The divergence is consistent with violation of the linear growth condition on the generator when the penalty gradient dominates execution revenue.

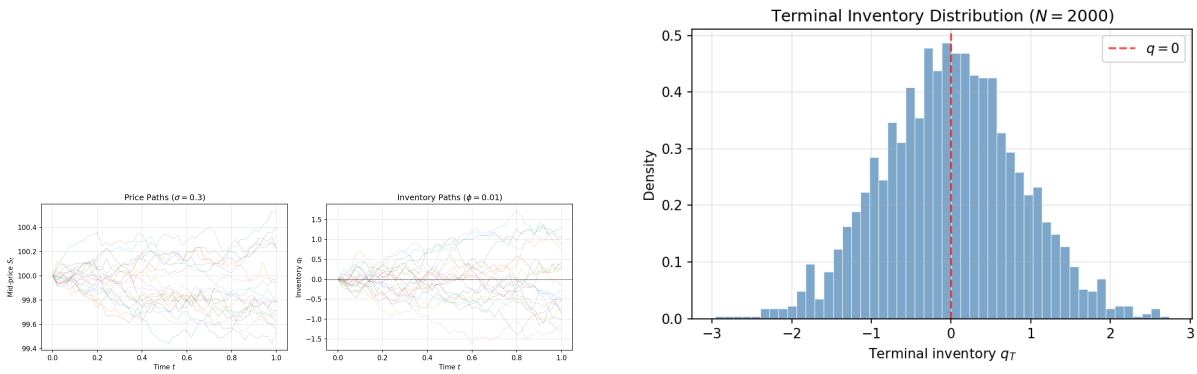
Non-linear penalties also widen the stationary FD optimal spread ($1.432 \to 1.483$), consistent with increased inventory aversion.

Table 5: Breaking point: parameter sweeps (3000 iterations, single seed).

Parameter	Value	Y_0	$\max\lvert Z_t\rvert$	Status
γ	0.5	0.460	0.05	Stable
	1.0	0.448	0.12	Stable
	2.0	0.349	2.22	Marginal
	3.0	-3.363	**258**	**Diverged**
	5.0	-17.96	58.7	Diverged
ϕ	0.01	0.456	0.09	Stable
	0.1	0.376	0.61	Stable
	0.5	0.185	1.48	Stable
	1.0	0.245	1.39	Marginal
	2.0	-6.051	**17.7**	**Diverged**

7.5 Forward Process Diagnostics

Figure 8 shows sample trajectories and terminal inventory distribution, confirming the forward SDE produces well-behaved paths with inventory centred near zero.



(a) Sample paths.

(b) Terminal inventory.

Figure 8: Forward SDE diagnostics (diffusion surrogate, $\phi = 0.01$).

7.6 Horizon Sensitivity

To empirically justify the finite-horizon truncation, we compare FD and BSDE at several horizons (Table 6).

Table 6: Horizon sensitivity: matched FD vs Deep BSDE surrogate (1000 iterations).

T	FD $V(0,0)$	BSDE Y_0	FD spread	Error
0.5	0.235	0.236	1.362	0.3%
1.0	0.457	0.457	1.370	0.1%
2.0	0.866	0.864	1.383	0.2%
5.0	1.852	1.172	1.410	36.7%

The solver is accurate ($< 0.3\%$) for $T \leq 2$ but fails at $T = 5$ (37% error). At $T = 5$ with $N_T = 50$, the step size $\Delta t = 0.1$ is large enough that the explicit Euler BSDE discretisation error accumulates significantly over 50 steps, and the training variance grows with the longer

trajectory. The FD spread converges toward the stationary value 1.432 as T increases, confirming the finite-horizon truncation is the source of the spread difference noted in Section 7.1.

7.7 Mean-Field Ablation

Comparing Type 1 (no coupling) and Type 3 (moment-proxy fictitious play) across 5 seeds, $Y_0 = 0.4566 \pm 0.0002$ vs 0.4568 ± 0.0001. The mean-field proxy has *no measurable effect* at $\phi = 0.01$, confirming that the current moment-based coupling is too coarse to influence the control in this parameter regime. Genuine distribution-dependent coupling is needed to make the mean-field interaction substantive.

8 Conclusion

We have proposed and empirically evaluated two deep BSDE solvers for optimal market-making in the Cont–Xiong dealer market model: a jump-aware formulation (FBSDEJ) preserving discrete Poisson inventory dynamics, and a diffusion surrogate for comparison. Against a matched finite-horizon finite-difference baseline, the surrogate achieves $< 0.1\%$ value error at $q = 0$ and its learned policy captures 99.3% of the FD-optimal P&L in forward simulation. However, the surrogate's structurally constant spread ($2/\alpha = 1.333$) differs from the FD spread (1.370) by 2.7%, demonstrating that *accurate value approximation does not guarantee accurate control recovery.* The jump model (2.4% value error at 20,000 iterations) achieves comparable loss to the surrogate (2.6×10^{-5} vs 3.4×10^{-5}) when given sufficient training, demonstrating that the initial accuracy gap was due to undertraining rather than a fundamental architectural limitation. Across 5 seeds, the jump model produces $Y_0 = 0.446 \pm 0.002$, confirming robustness. The jump formulation preserves the discrete control structure at the cost of slower convergence. Notably, the cross-dynamics evaluation shows that the surrogate-trained policy remains economically robust when deployed under true Poisson execution — in low inventory-penalty market-making regimes, errors introduced by diffusion-based approximations may have limited economic impact despite inducing structurally incorrect controls. This robustness should not be assumed in models with stronger inventory penalties or adverse selection.

Limitations. The experiments are at $d = 2$ (effectively $d = 1$ after the reduction in Proposition 2) where finite differences are superior. The mean-field coupling is moment-based and, as the ablation shows, currently inactive at the tested parameters. The stress tests use 5 seeds and 3000 iterations. The terminal condition and truncation are heuristic.

Future work. Distribution-dependent mean-field coupling via particle systems and empirical measure encoders; common noise via deep signatures Hu et al. [2025]; Nash vs Pareto comparison Cont and Xiong [2024]; higher-dimensional multi-asset formulations; systematic instability characterisation with more seeds and longer training.

Code. `https://github.com/cgarryZA/DeepBSDE`

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